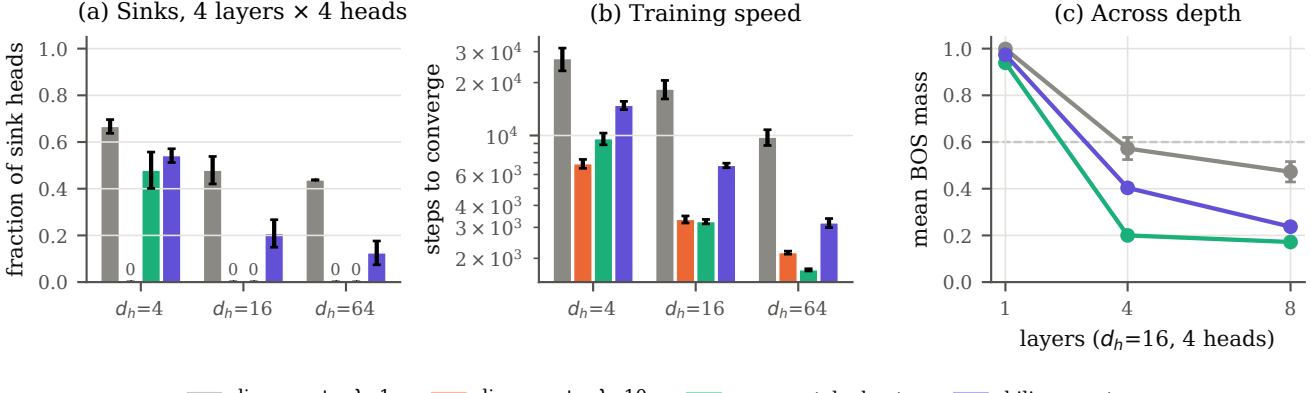
(a) Sinks, 4 layers × 4 heads
fraction of sink heads
1.0
0.8
0.6
0.4
0.2
0.0
0
0 0
0 0
d_h=4
d_h=16
d_h=64
(b) Training speed
steps to converge
3×10^4
2×10^4
10^4
6×10^3
4×10^3
3×10^3
2×10^3
d_h=4
d_h=16
d_h=64
(c) Across depth
mean BOS mass
1.0
0.8
0.6
0.4
0.2
0.0
1
4
8
layers (d_h=16, 4 heads)
linear gate, λ=1
linear gate, λ=10
race-matched gate
bilinear gate